

Confidence Calibration in Ordinal Classification: A Comparative Study on Oil Palm Fruit Ripeness

By Afonso CRUZ, Ayrton CRUZ, and Tomas KULZER

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1. Introduction

This report presents our final project for the Machine Learning curricular unit which consisted of an automated classification system for Oil Palm Fruit Quality Grading with reliable confidence estimation. We start with an introduction of our work and move on to the methodology, listing and explaining the different algorithms and methods used. We conclude our report by showing and discussing our results and with a brief conclusion.

1.1. Problem and Motivation

In many applications, outputting not only the predicted class but also a reliable confidence score for that prediction is crucial, particularly for ordinal classification tasks where classes follow a natural order. This project follows Proposal 2, which tasks us with studying confidence estimation techniques for ordinal data classification, measuring calibration error, and exploring methods to minimize it.

1.2. Objectives

The main objectives of this project were:

- To develop an ordinal classification system for oil palm fruit ripeness.
- To implement and compare multiple confidence calibration methods (Platt Scaling, Temperature Scaling, Isotonic Regression).
- To evaluate calibration performance using metrics such as Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and Negative Log-Likelihood (NLL).

1.3. Approach Overview

We approached this project with an educational mindset, meaning that instead of trying to blindly obtain the best possible results in terms of metrics, we focused on understanding the concepts and implementing (sometimes inferior) methods that were simpler. One example of this was our feature extraction pipeline where we used traditional algorithms instead of deep learning.

2. Methodology

2.1. Dataset

The dataset chosen was obtained from the Mendeley Data website and comprises 4,728 high-resolution images of oil palm

Class	N° Samples	%
Immature - 0	758	16.03
Partially Ripe - 1	1405	29.72
Fully Ripe - 2	1489	31.49
Overripe - 3	665	14.07
Decayed - 4	411	8.69

Table 1. Dataset sample distribution

fruits across five ordinal ripeness levels: Immature, Partially Ripe, Fully Ripe, Overripe, and Decayed [1]. Captured under real-world conditions in Central Kalimantan, Indonesia, it captures environmental variability including diverse lighting, poses, and backgrounds from multiple devices.

Initially, to facilitate efficient experimentation and reduce computational demands during development, a balanced subset of 300 images per class (1,500 total) was used by randomly selecting 300 images per class using a Python script for training, validation, and testing. This approach enabled rapid iteration on feature extraction, model training, and calibration pipelines. However, upon training with lighter and faster models, like lightGBM, we realized we could use the full dataset and still have short training times so our final implementation uses the full dataset with 4,728 images.

It is worth noting that the dataset class samples are not balanced and are distributed as shown in Table 1.



Fig. 1. Dataset Sample Grid - 5 Classes x 5 Samples

2.2. Data Loading and Preprocessing

We implemented a data loading and preprocessing pipeline that scans the dataset directory for all class folders (e.g., 0Immature, 1PartiallyRipe, etc.), assigns ordinal labels, splits the data and generates augmented images.

2.2.1. Data Loading

Each image is loaded and converted to a 256x256x3 (RGB) numpy array to ensure ease of processing.

2.2.2. Ordinal Label Encoding

Folder names provide explicit ordinal mapping: "0Immature" → 0, "1PartiallyRipe" → 1, "2FullyRipe" → 2, "3Overripe" → 3, "4Decayed" → 4, preserving the natural ordered progression of ripeness stages. This integer encoding (0-4) enables classifiers like SVM, Random Forest, and Gradient Boosting to leverage ordering constraints rather than treating classes as categorical.

2.2.3. Data Splitting

The dataset undergoes stratified splitting to maintain class distribution across sets: 60% training, 20% validation, and 20% testing, performed by our `split_dataset.py` script.

2.2.4. Data Augmentation

Data Augmentation is a very useful way of cheaply enlarging the training set. It works by applying label-preserving transformations to copies of samples. These transformations can be rotations, flips, noise and help make the model more robust to common variations in real-world data.

2.3. Feature Extraction

Although feature extraction using deep learning approaches usually yields better results, we opted for a more traditional approach using 3 different feature extraction algorithms and then combining the resulting features.

2.3.1. HOG

HOG (Histogram of Oriented Gradients) extraction works by computing gradient magnitudes and orientations on grayscale images. We hoped using HOG extraction would capture fruit edges and textures that could then be useful for classification.

2.3.2. SIFT

Scale-Invariant Feature Transform detects keypoints related to different shapes independently of their scale. In our implementation, we used SIFT to capture ripeness-specific texture patterns.

2.3.3. Color Histograms

Color histograms capture the global color distribution which is critical for oil palm ripeness classification. In our implementation, we utilized the `ColorHistogramExtractor` class to process images from the dataset splits, converting them into the HSV color space to mitigate lighting variations. For each image, we computed a 32-bin histogram for each of the three channels

(Hue, Saturation, and Value), which are subsequently concatenated to form a 96-dimensional feature vector. These feature vectors are then aggregated with their corresponding labels and serialized into joblib files to serve as inputs for the classification models.

2.3.4. Feature Combine

After all feature extraction algorithms are concluded we need to combine them so they can then be feed to the classifier models. Our `feature_combiner.py` script concatenates the multiple feature types (HOG, SIFT, color histogram) into a single feature vector for each image, then normalizes them using `StandardScaler`.

2.4. Classification Approach

For classification, we used three different models: SVM, RF and GB.

2.4.1. SVM

Support Vector Machines (SVM) classify data by constructing optimal decision boundaries that separate the different classes. This algorithm focuses on maximizing the distance (margin) between these boundaries and the nearest data points of each class to ensure accurate predictions. For our implementation, we adopted a threshold-based ordinal approach to respect the natural order of the fruit stages. Specifically, for the 5 ripening classes, the model trains 4 internal binary classifiers, where each one determines if a fruit surpasses a specific maturity threshold (for example distinguishing 'Immature' from all later stages), combining these results for the final classification.

2.4.2. Random Forest

Random Forest is an ensemble learning method that builds multiple decision trees to create a single, robust model. For our implementation, we utilized a direct multiclass approach rather than a specific ordinal architecture. To ensure high performance, we applied strict regularization parameters (such as limiting tree depth and setting a minimum number of samples per leaf) to prevent overfitting and ensure the model generalizes well to new fruit images. For the implementation of both the Support Vector Machine (SVM) and Random Forest models, we utilized the scikit-learn framework.

2.4.3. Gradient Boosting

Gradient Boosting works by optimizing many weak models and combining them to create a single strong model.

For our implementation, we used the gradient boosting framework `LightGBM` [2]. It was designed to have faster training speed, higher efficiency, lower memory usage and better accuracy using tree based learning algorithms.

2.5. Confidence Estimation and Calibration

2.5.1. Important Definitions

Logits are the raw, unnormalized outputs of a classification model. These logits (z_i) are transformed via the **softmax** function (1).

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (1)$$

Softmax converts raw logits into probabilities, serving as the baseline confidence estimation. While softmax produces valid probability distributions, these confidences often suffer from miscalibration which is when predicted confidence levels don't align with actual accuracy rates. This is the uncalibrated baseline that we then compare calibration methods to.

Reliability diagrams (ex. Figure 2) plot the accuracy of a model as a function of its confidence output and are a visual representation of its calibration. [3]

Expected Calibration Error (ECE) divides predictions into confidence bins and, for each bin (B_m), compares the average accuracy ($acc(B_m)$) with the average confidence ($conf(B_m)$). ECE is the weighted average of the absolute differences across bins. [3]

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |acc(B_m) - conf(B_m)| \quad (2)$$

Maximum Calibration Error (MCE) measures the worst-case deviation between confidence and accuracy across bins. [3]

$$MCE = \max_{m=1, \dots, M} |acc(B_m) - conf(B_m)| \quad (3)$$

Negative Log Likelihood (NLL) measures how well predicted probabilities match true labels by penalizing low probability assignments to correct classes. [3]

$$NLL = -\frac{1}{N} \sum_{i=1}^N \log(p_{y_i}) \quad (4)$$

p_{y_i} is the probability of the correct class for sample i .

Lower NLL means better-calibrated probabilities.

2.5.2. Calibration Methods

Platt scaling is a parametric probability calibration method that fits a logistic regression model to the raw outputs of a classifier using a validation set. [4] Originally designed for binary SVMs, it transforms uncalibrated scores $f(x)$ into calibrated probabilities via the sigmoid function:

$$P(y = 1|f(x)) = \frac{1}{1 + \exp(A \cdot f(x) + B)} \quad (5)$$

where A and B are parameters learned by minimizing NLL on calibration data.

Temperature Scaling is the simplest extension of Platt Scaling and uses a single scalar parameter (T) for all classes. This technique applies the modified softmax:

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (6)$$

where T minimizes NLL on a validation set. Since T does not affect the maximum of the softmax function, "temperature scaling does not affect the model's accuracy" [3].

Isotonic Regression is a flexible non-parametric calibration technique that fits a monotonically increasing curve to map raw classifier outputs (logits) to calibrated probabilities. Unlike the two other calibration methods mentioned earlier, isotonic regression aims to reduce MSE (Mean-Squared Error) which is calculated in the following way [5]:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7)$$

3. Results

3.1. Classification Performance and evaluation

Initially, we used the 3 different feature extraction algorithms mentioned earlier, however, when testing different combinations of features, we found the best accuracy values were found when using only color histograms. This suggests that color carries significantly more importance than texture or shape for oil palm fruit ripeness classification so all our final models were trained using only color histograms. Comparing models was straightforward, since we only had 3 and GB had an accuracy of approximately 88% compared to the approximately 81% and 63% of the other models. However, when dealing with ordinal classification, accuracy doesn't always tell the full story, since a classifier that makes many small errors could be better than a classifier that makes fewer, larger errors. To address this, ordinal classifiers are usually also evaluated using MAE (Mean Absolute Error) and MSE (Mean Squared Error) [5]. In regards to these metrics: "MSE is better in situations where the severity of the errors is more important, while MAE is better in situations where the tolerance for small errors is lower" [6] because MSE penalizes large errors more heavily due to squaring and MAE treats all errors linearly via absolute values. MSE's formula (7) was mentioned in the last section and is similar to MAE's formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (8)$$

Table 2 presents the model evaluation metrics results obtained using the validation set which solidify GB as our best performing model, followed by RF.

Model	Accuracy	MAE	MSE
GB	0.879	0.169	0.281
RF	0.808	0.291	0.519
SVM	0.626	0.494	0.785

Table 2. Final Model evaluation metrics (validation set)

3.2. Calibration Results

To compare the different calibration methods we started by applying the Softmax function to serve as a confidence estimation baseline for the rest of the methods. For each method we computed the resulting NLL, MCE and ECE, shown in Table 3.

Model	Calibration	NLL	MCE	ECE
GB	Baseline (Softmax)	0.429	0.187	0.0486
GB	Platt Scaling	0.405	0.388	0.0147
GB	Temperature Scaling	0.412	0.287	0.0242
GB	Isotonic Regression	0.640	0.387	0.0326
RF	Baseline (Softmax)	0.718	0.288	0.1952
RF	Platt Scaling	0.573	0.193	0.0401
RF	Temperature Scaling	0.591	0.261	0.0452
RF	Isotonic Regression	0.905	0.373	0.0292
SVM	Baseline (Softmax)	1.028	0.0719	0.0467
SVM	Platt Scaling	1.027	0.125	0.0802
SVM	Temperature Scaling	1.017	0.308	0.0517
SVM	Isotonic Regression	1.070	0.674	0.0499

Table 3. Calibration metrics for different models with different methods (with $M = 15$ bins)(test set)

4. Discussion

4.1. Key Findings

For our three classification models, we found out that none of the three calibration methods used stood out as the universal best. Instead, each classifier had its own best performer. GB had the lowest ECE with Platt Scaling, RF’s ECE benefited the most from Isotonic Regression while our SVM’s lowest ECE was actually using the baseline Softmax.

4.2. Interpretation

4.2.1. Gradient Boosting

Figure 2 shows the different reliability diagrams of the Gradient Boosting classifier when uncalibrated and when calibrated with the 3 implemented methods. The diagrams are consistent with the results in Table 3, showing Platt Scaling as the best performer in terms of ECE, while having the highest MCE. The NLL values of Platt and Temperature Scaling are lower than the baseline, as expected, since that is the metric they minimize. Isotonic Regression stands out with a higher NLL than even the baseline since this method minimizes MSE, unlike the two others.

It should be noted that the methods use the validation set to minimize their respective metrics, meaning these won’t necessarily be minimized on the test set which was the one used to measure the results.

4.2.2. Random Forest

Looking at the results for Random Forest (Figure 3) we identify a notably under-confident classifier. With this model, the lowest ECE is achieved using Isotonic Regression although this method increases NLL relatively to the baseline, likely due to overfitting on the limited validation data. Once again we see Platt and Temperature Scaling having the lowest NLL values.

4.2.3. SVM

Our SVM classifier, which had the lowest accuracy out of the 3 models, boasts a remarkably calibrated baseline (Figure 4), with the lowest ECE and MCE of the 3 baselines. This reflects SVM’s decision function outputs that naturally produce well-spread probabilities after softmax in this 4-binary-threshold or-

dinal setup. Platt and Temperature Scaling both manage to reduce NLL at the cost of increasing both ECE and MCE. Isotonic Regression severely harms all metrics, especially MCE (9x worse), once again probably due to non-parametric overfitting on sparse validation data.

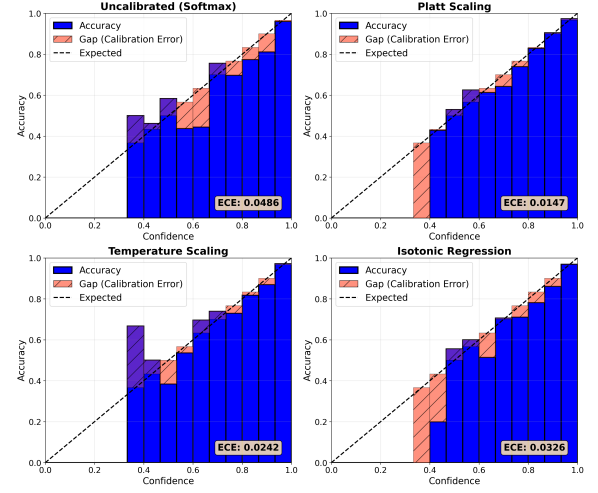


Fig. 2. GB Reliability diagrams

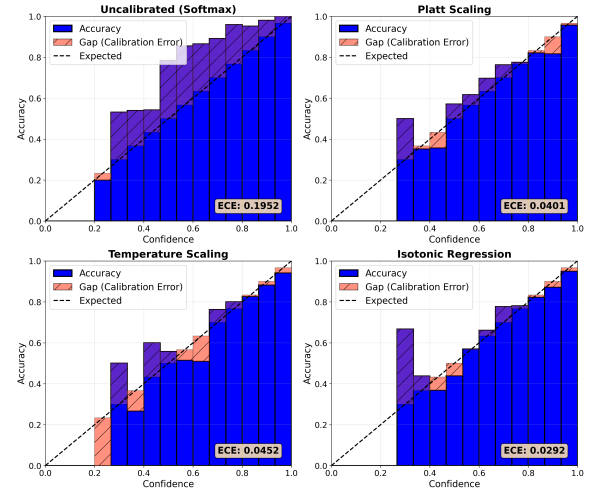


Fig. 3. RF Reliability diagrams

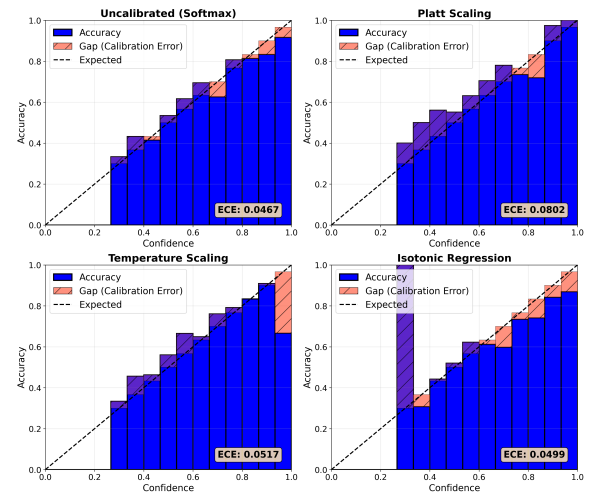


Fig. 4. SVM Reliability diagrams

4.3. Practical Implications

Given that Gradient Boosting achieved approximately 88% accuracy with the lowest Expected Calibration Error (ECE = 0.0147) using Platt Scaling, the model demonstrates possibly sufficient reliability for commercial fruit sorting systems. The calibrated confidence scores would enable confidence-aware decision making for borderline predictions: where confidence is low, the system could automatically defer to human inspection, reducing both false rejections and misclassifications that would impact yield and product quality. The most significant practical implication of our calibration results is the suggestion that calibration should be model specific which justifies investing resources in cross-validation with multiple calibration strategies.

4.4. Limitations

4.4.1. Dataset Limitations

The fact that the dataset we used consists of pictures only from Central Kalimantan, Indonesia, means that our system would probably not work across different regions since ripeness indicators could vary significantly across geographic locations and seasons.

4.4.2. Feature Extraction

Our decision to use traditional feature extraction algorithms (HOG, SIFT, Color Histograms) in favour of modern deep learning approaches meant that we were possibly leaving accuracy gains on the table.

4.4.3. Ordinal Constraint Underutilization

Of our three classifiers, only SVM has an explicitly enforced ordinal structure while Random Forest and lightGBM treat the problem as standard multiclass classification, which means these are potentially discarding valuable information about class similarity. We initially explored variants of both RF and GB that were ordinal oriented but they resulted in less accuracy while being more computationally intensive most of the time.

4.4.4. Metric Selection

Our calibration evaluation focused on ECE, MCE and NLL, which although appropriate, fail to capture dataset and business specific costs. For ripeness classification, misclassifying "Fully Ripe" as "Decayed" carries different economic consequences than misclassifying "Immature" as "Fully Ripe," but the selected metrics treat errors uniformly. Custom loss functions weighted by economic impact would provide better decision support for eventual deployment.

4.4.5. Computation Cost

Since our dataset was not particularly heavy and the models used had short training times, we didn't measure any benchmarks which would be critical constraints with different datasets/models or for real-time sorting line integrations.

5. Conclusion

During this project, we successfully developed an ordinal classification system for oil palm fruit ripeness with calibrated confidence estimation. Using traditional feature extraction methods (initially 3) we were able to train 3 different classifiers which were then used to compare different confidence calibration methods. Critically, our results showed that confidence calibration effectiveness is model dependent: Platt Scaling optimized Gradient Boosting's calibration (ECE = 0.0147), Isotonic Regression best served Random Forest (ECE = 0.0292) and the baseline softmax proved surprisingly effective for SVM (ECE = 0.0467). From an educational point of view, this project forced us to implement a complete classification system which allowed us to learn about a lot of models, algorithms and methods which we probably would never hear of otherwise.

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CRedit Author Statement

Afonso CRUZ: Software, Writing, Visualization, Validation	- 33.3%
Ayrton CRUZ: Writing, Reviewing and Editing	- 33.3%
Tomas KULZER: Conceptualization, Methodology, Software, Investigation, Writing	- 33.4%